

Exploring the prognostic potential of let-7 microRNA in clear cell renal cancer

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15 SUMMARY

16 Kidney cancer remains one of the most widespread strains of cancer, with clear cell renal
17 cell carcinoma (ccRCC) being among the most prevalent, aggressive, and deadly forms.
18 Despite its prevalence, early, non-invasive detection remains a challenge. MicroRNAs
19 (miRNAs), including the let-7 family, are small non-coding RNAs, that frequently show
20 abnormal regulation in cancer. However, existing studies disagree on whether let-7
21 expression increases or decreases in ccRCC. To address this gap, we examined whether
22 let-7 is positively or negatively correlated with ccRCC and identified which let-7 miRNA might
23 serve as the most effective biomarker. We hypothesized that let-7 would generally decrease
24 expression in ccRCC, based on its behavior in other cancers. We also hypothesized that let-
25 7a or let-7b would be the most promising biomarkers based on their detectability in urine
26 supernatant and other cancers. We applied statistical analyses to patient data from the
27 Genomic Data Commons to identify direction and significance of regulation. We also trained
28 a machine learning model on let-7b expression to investigate its potential for early, non-
29 invasive ccRCC detection. Unexpectedly, we documented an increase in let-7 expression in
30 ccRCC, contradicting the prevailing understanding and our hypothesis. This suggests that
31 the role of let-7 in ccRCC may differ from its role in other cancers. Despite this unexpected
32 result, let-7b demonstrated the most significant expression difference between normal and
33 cancerous tissue, especially in the early stages. The machine learning model trained on let-
34 7b expression achieved high classification accuracy, showing promise for early, non-invasive
35 ccRCC detection. These findings contribute to resolving existing conflicts in the literature and
36 highlight let-7b's potential as a diagnostic biomarker.

38 INTRODUCTION

39 Worldwide, approximately 263,500 instances of clear cell renal cell carcinoma (ccRCC) are
40 diagnosed annually, making it the most prevalent subtype of renal cell carcinoma (RCC) (1).
41 RCC accounts for more than 90% of all kidney cancer cases, and ccRCC constitutes 75–
42 80% of RCC cases (1). Early-stage ccRCC is asymptomatic and can remain undetected until
43 it has already progressed to an advanced stage (1). It is a primary tumor that begins in the
44 kidneys and does not originate from the spread of other organs. It can be hereditary, but it
45 usually stems from unknown causes. Diagnosis often begins with imaging techniques such
46 as computed tomography (CT) scans or ultrasounds, followed by a biopsy or magnetic
47 resonance imaging (MRI) (2).

48 The conventional diagnostic approaches are limited in several ways. Firstly, CT scans and
49 ultrasounds may not always be able to distinguish kidney masses from cancer. They are

also limited in their sensitivity to small tumors. Ultrasounds are also operator dependent, thus making them less reliable. Biopsies have a chance of yielding a non-diagnostic result; 10–20% of kidney biopsies do not yield a definitive diagnosis, resulting in a longer delay and potentially requiring additional biopsies. All these limitations can lead to a lowered prognosis for patients with ccRCC (3–5).

Early and accurate cancer detection is critical for improving patient outcomes, underscoring the need for reliable biomarkers (6). This is where microRNAs (miRNAs) emerge as potential solutions. MiRNAs are short non-coding RNAs that regulate gene expression by inhibiting protein production from target messenger RNAs (mRNAs), an intermediary RNA that carries genetic information from DNA to the ribosome for protein synthesis. The disabling of mRNA molecules is done through the RNA-Induced Silencing Complex (RISC). RISC components, such as Argonaute proteins, bind miRNAs to target specific mRNAs through sequence complementarity, primarily between the 3' Untranslated Region (3'UTR) of the mRNA and the miRNA seed region with imperfect base pairing (8). MiRNAs can silence multiple mRNAs by preventing ribosome attachment or promoting mRNA degradation. This inactivation of mRNAs results in a post-transcriptional silencing to effectively control or fully deactivate specific proteins. The miRNAs produced by the cell, and thus the silenced proteins, differ significantly between healthy and diseased tissues (7).

Some miRNAs act as tumor suppressors, silencing genes that promote cell proliferation and inhibiting a cancer's growth and maturation (9). They are usually underexpressed in higher stages of cancers. Other miRNAs function as oncogenes by silencing genes that inhibit proliferation, driving the progression of cancer (9). Thus, they are usually overexpressed in higher stages of cancers (9). Therefore, the aforementioned dysregulation of miRNAs has been consistently shown to contribute to cancer development (9).

One of the most researched miRNAs is the let-7 family of miRNAs and is also well established as a tumor suppressor (10). Let-7 is known to inhibit many oncogenes that support cell proliferation and cancer stem cell (CSC) behavior, such as Kirsten rat sarcoma viral oncogene homolog, cellular myelocytomatosis oncogene (c-MYC), High Mobility Group AT-hook 2 (HMGA-2), and others (10).

Specifically in kidney physiology, let-7 is known for regulating normal renal function by controlling the progression of the cell cycle and tissue homeostasis (11). For example, it is known to suppress oncogenes and fibrotic pathways (those that drive scar tissue formation), such as the TGF- β signaling pathway (12, 13). Let-7 is also known to be involved in

nephrogenesis, with higher levels signaling the termination of nephrogenesis (14). Although many let-7 members share common seed regions and overlapping biological functions, individual members often also have their own niches and may have abnormalities in regard to their regulation in cancer (13). In particular, let-7b's regulation has been shown to be context dependent (10, 13). This distinctive behaviour suggests that let-7b has a unique regulatory role in cancer.

In ccRCC, let-7 is generally known to be downregulated, supporting its widely accepted tumor-suppressive role (13). Gowrishankar et al. looked at a total of 99 samples and found that most let-7 members were downregulated compared to adjacent normal tissue, with let-7b approaching statistically significant downregulation (9). Interestingly, they observed that let-7g, let-7i, and let-7f were upregulated compared to the normal tissue (9). In general, the study documented the status quo, suggesting that let-7 is a tumor suppressor.

However, there was a study by Fedorko et al. that showed contrasting results (11). In that study, urine supernatant from kidneys with ccRCC was analyzed, and all six let-7 family members examined were found to have increased expression (11). While not directly comparable and not disproving one another per se, these studies nonetheless raise questions on let-7's behaviour within the kidney tissues of ccRCC patients. To further clarify this uncertainty, it warrants more study to use tissue-level data to classify the regulation of let-7 members.

To address this research gap, we analyzed expression data from kidney tissue by using the Cancer Genome Atlas (TCGA) project from the Genomic Data Commons (GDC). Specifically, the samples among the KIRC cohort were used, referred to as TCGA-KIRC. We hypothesized that let-7 expression would decrease in the samples from ccRCC patients, as was typical in most studies (10). We also hypothesized that let-7a and let-7b would exhibit the largest differences in expression, as similar results were reported in a study analyzing urine supernatant in ccRCC as well as other studies in other cancers (11, 15, 16).

To test our hypotheses, we extracted the expression from all let-7 members from the TCGA-KIRC project in the GDC and conducted an independent two-sampled t-test to see the trend and the significance of the regulation. We found that let-7 expression overall increased, with only let-7c and let-7e having decreased. Notably, let-7b showed the largest difference in expression between normal and cancerous tissue, supporting its potential as a biomarker. This finding implies that in a clinical setting, the best let-7 miRNA to detect ccRCC may be let-7b. Early detection using miRNA biomarkers like let-7b could improve prognosis for ccRCC patients by offering a less invasive and more accurate diagnostic tool.

RESULTS

Let-7 has a well-established association with ccRCC, and we explored its use for earlier diagnosis of ccRCC (13). We aimed to determine if let-7 would be downregulated or upregulated overall. To evaluate this, we quantified the expression of let-7 with the unit “reads per million miRNA mapped” (RPM). This adjusts the raw count (the number of times a particular miRNA is detected) for the total reads in the sample.

We obtained the miRNA concentration data from the TCGA-KIRC dataset and graphed the data of the let-7 members across the stages of cancer. There were 20 normal samples and a total of 537 cancer samples spread across the four stages of cancer. We investigated all available let-7 members and presented their expression levels. We determined the significance of the differences between normal and cancer samples by obtaining the p-value for each let-7 member with a two-sample independent t-test.

For a more fine-grained understanding of let-7 miRNAs’ stage-dependent variations, we plotted the average expression level versus cancer stage to identify stage-specific variations in let-7 expression and pinpoint potential thresholds where expression levels significantly change. We see that let-7b expression minimally varied between the cancer stages but increased dramatically from the normal to cancer stages (**Figure 1**). To summarize the results and the trends in the data, a violin plot was constructed, representing all 11 let-7 miRNAs tested and their varying expression (**Figure 2**).

On the application of the t-test, the majority of let-7 family members, like let-7a, showed a slight increase in cancer samples with a p-value approaching statistical significance (**Figure 2**). Let-7f, let-7g, and let-7i saw a statistically meaningful increase. Let-7c showed a decrease approaching statistical significance. However, this result was likely due to the presence of three large outliers, which can be seen in the let-7c scatterplot, and was evaluated using the z-score (**Appendix**). Let-7d saw relatively no change, and let-7e saw a statistically significant decrease. Out of all let-7 miRNAs tested, let-7b had the largest difference between the normal and cancerous samples, with the lowest p-value ($p = 1.18 \times 10^{-7}$). We found that across all stages of ccRCC, let-7b’s expression was significantly different compared to the normal tissues. Comparing adjacent groups, there was a highly significant difference between normal tissues and Stage 1 of ccRCC, a difference approaching statistical significance between Stage 1 and Stage 2, a moderately significant difference between Stage 2 and Stage 3, and no statistically significant difference between Stage 3 and Stage 4 (**Table 1**). The above results indicated that let-7b would be the best biomarker out of the let-7 family.

To demonstrate this, we constructed a Support Vector Machine (SVM) model in Python to predict the tissue status (cancerous or normal) based on the expression level of let-7b. An SVM model is a supervised predictive machine learning model used in research to differentiate between multiple classes for predictive purposes. We created an SVM model using the scikit-learn (or sklearn) library, a popular library for machine learning in Python. The SVM model was implemented using the Support Vector Classifier (SVC) in scikit-learn. In order to mitigate the present class imbalance in the data, which could lead to misleading metrics and poor generalizability, we applied oversampling. Oversampling, or the artificial and intelligent addition of samples to the minority class, was employed using the technique known as Synthetic Minority Oversampling Technique (SMOTE).

Upon training and oversampling, we evaluated and tabulated the results and metrics of our model (**Table 2**). Relevant metrics for evaluation include accuracy, precision, recall, and Receiver Operating Characteristic-Area Under Curve (ROC-AUC). Accuracy, or the ratio of correct over total predictions, indicates how often the SVM model assigns the correct class; however, it is subject to overfitting biases. Precision is the accuracy over only the positive class (cancerous). Recall (or sensitivity) is the accuracy over only members of the negative class (normal). ROC-AUC gives the probability that the SVM model ranks a random positive sample higher than a random negative sample. ROC-AUC's primary advantages are that it accounts for class imbalance and it quantifies separation ability.

In the classification report for the SVM, evaluated on a dataset of 215 samples, the model predicted 195 samples correctly and only 20 samples incorrectly. The SVM had a precision of 0.93 and a recall of 0.88 for normal samples. Furthermore, for cancerous samples, the model had a precision of 0.89 and a higher recall of 0.94 (**Table 3**). The overall accuracy of the model was 0.91, indicating that 91% of all predictions were correct (**Table 3**). The harmonic mean of the precision and recall is defined as the F1 score. This statistic gives a balanced look at the two ways of measurement. Both macro and weighted averages for F1 score, recall, and precision are equal to 0.91, suggesting balanced performance across both classes. The model shows slightly better recall for cancerous samples and better precision for normal samples (**Table 3**).

DISCUSSION

Let-7 is known for its potential diagnostic purposes (10). We aimed to determine whether let-7 would be upregulated or downregulated in ccRCC. The analysis showed that, between

normal and cancerous tissues, the expression of let-7 members mainly increased (e.g. let-7b), while some decreased (e.g., let-7c), and others were not correlated with ccRCC (e.g., let-7d). The let-7 member with the largest difference between normal and cancerous tissues was let-7b.

For a model to be effective at diagnosing a tissue as cancerous, the separation between normal and cancer classes should be large. Additionally, the concentration should be adequately high (in the thousands of RPM) for the miRNA to be easier to detect and distinguish in sequencing data. This is because the higher the expression level, the more frequently the sequence occurs in the sequencing data, increasing the signal-to-noise ratio (18). A signal-to-noise ratio measures how strong the actual signal (desired sequence) is compared to the background noise. The high signal-to-noise ratio stemming from the high concentration of the desired sequence makes it easier to detect from the background noise. This facilitates confident statistical analysis and robust identification, which is critical for both research and clinical applications.

The TCGA-KIRC database used in the present study was used in earlier studies that showed that high expression of let-7 members correlated with lower survival for ccRCC, and vice versa (10, 19). This was evaluated using a tool called Prognostic MicroRNA Tool (PROGmiR) (10, 19). PROGmiR is designed to help researchers investigate the prognostic significance of miRNAs across various cancer types using publicly available data sets. The study used the survey of all 33 human cancer databases from TCGA and Gene Expression Omnibus (GEO) that had been uploaded to PROGmiR and showed that across all cancers, the upregulation of the majority of let-7 members was significantly associated with better prognosis, supporting its generally accepted tumor-suppressive role (10, 19). In ccRCC specifically, all let-7 members were positively correlated with prognosis. The only exceptions to this statement were let-7g and let-7f, which showed no statistically significant correlation, and let-7i, which was negatively correlated with prognosis. These observations suggested that let-7 is a tumor suppressor in most cancers, including ccRCC. PROGmiR does not train a prediction algorithm from scratch but rather analyzes existing survival data using standard statistical methods like Cox proportional hazards models and Kaplan-Meier survival curves (19). This allows researchers to form hypotheses about the tumor-suppressive or oncogenic roles miRNAs may play in cancer (19).

While both the PROGmiR study and ours used the TCGA-KIRC dataset, the structure of the data used, the questions asked, and the analysis methods were different. PROGmiR only analyzed samples from cancer patients to determine whether higher or lower levels of let-7

were associated with better prognosis (19). In contrast, we analyzed expression data from both normal and cancerous tissue types to identify whether let-7 members were upregulated or downregulated in ccRCC (10, 17). This direct comparison between healthy and tumor tissue allowed us to assess differential expression more clearly. In addition, our methodology was relatively simple—two clearly defined sample types (normal vs. cancerous) were directly compared to determine expression changes—whereas PROGmiR's predictive survival analysis was built on statistical models within the cancer-only group (19).

This distinction is important. Our study focused on resolving whether let-7 is upregulated or downregulated in ccRCC and determining which let-7 member has the greatest biomarker potential for detecting ccRCC. Their study focused on patient survival, using only cancer samples and Kaplan-Meier-based statistical predictions, with the goal of assessing whether relative differences in miRNA expression levels among cancer patients correlated with clinical outcomes (19). Their scope was the whole tumor and patient survival; ours was the tumor versus normal comparison to identify expression patterns (10, 19).

Therefore, while our findings might seem to conflict with PROGmiR's, they are answering different questions. For example, we found that let-7b might be upregulated in cancer tissue compared to normal tissue, and PROGmiR showed that, among cancer patients, those with even higher expression of let-7b might have better survival (19). These two outcomes are not contradictory and could be complementary. Our goal was to classify tissue as cancerous or not based on the expression of let-7 members, while theirs was to predict patient survival based on relative miRNA levels within the cancer group (19). These differences in sample subsets, analytical approach, scope, and interpretation explain why both studies can use the same dataset and still arrive at different—but not mutually exclusive—conclusions. One potential explanation could be a compensatory tumor-suppressor response. The development of cancer creates oncogenic stress, which may result in an increased counter-regulatory expression of tumor suppressors by the cell to counteract the development of cancer (20). This would explain why certain tumors have upregulated tumor suppressors that may be involved in such pathways. However, with the artificial addition of tumor suppressors, it is able to mitigate and perhaps reverse the effects of cancer, consistent with PROGmiR's predictions.

Our initial hypothesis was that a decrease in let-7 members would occur overall, as let-7 is a tumor suppressor and this result was observed in many other papers (10, 17–19, 21, 22). However, this hypothesis was not borne out. There was a 2.63-fold increase in expression of let-7b between the normal tissue and the cancerous stages on average. This suggests that

in ccRCC, let-7b might act as an oncogene—an uncommon idea. We found let-7b to have the lowest p-value and most significant increase; however, let-7f, let-7g, and let-7i showed highly significant increases as well. This result is consistent with the findings of Fedorko et al. (11). However, in our study, let-7a showed a decrease approaching statistical significance, while let-7e showed a highly statistically significant decrease, with the other miRNAs showing statistically insignificant regulations. These results diverge from what was observed in the urine, where Fedorko et al. observed an increase in all let-7 members. This highlights the need for further investigation into an adequate explanation of expression among different members of the let-7 group.

tumorA review by Boyerinas et al. documented that let-7b was upregulated in gastrointestinal stromal tumors (GIST) and in lymphoma (23). In lymphoma specifically, Lawrie et al. found let-7b elevated in diffuse large B-cell lymphoma (DLBCL) and follicular lymphoma cases of the lymph nodes that subsequently underwent high-grade transformation to DLBCL (24). This finding is consistent with the possibility that let-7b promotes cancer progression. Elevated let-7b expression has also been detected in the urine supernatant of ccRCC patients as previously discussed, matching our tissue-level finds (11). The authors proposed that tumor cells may passively shed this miRNA more, though the mechanism was not specified. In blood cancers, Yazarlou et al. reported let-7b upregulation in AML bone marrow compared to normal controls (25). Collectively, let-7b upregulation has been demonstrated in kidney tissue, B-cell lymph nodes, soft tissue, and bone marrow, challenging the current tumor-suppressive notion and suggesting a more context-dependent role of let-7 members. The mechanisms causing this variability remain to be definitively described, and a more systematic survey across cancer types would be beneficial to determine whether the upregulation observed in ccRCC is truly cancer-dependent or a shared phenomenon.

We have also shown let-7b to be a promising candidate for a model that identifies a tissue as cancerous or not. Since the p-value was far below the threshold, a model based on the let-7b expression is likely to perform very well. The SVM model we constructed had a high accuracy, F1 score, and ROC-AUC score, with all benchmarks approximately 0.9, showing comparable and slightly better performance than a similar SVM model made by Li et al. (Table 3) (27). Their model used 23 unnamed genes, unlike our sole usage of let-7b, to also predict stages of ccRCC samples (27). They had more samples than our study (604 compared to our total of 557 samples) and achieved an accuracy of 81.1% and an AUC of 0.86, comparable to our 90.6% accuracy and 0.93 ROC-AUC score (Tables 2, 3) (27). This suggests that let-7b is a viable biomarker for cancer with potential for more advanced neural

network study in the future, and could allow the classification of tissues using let-7b to be a reliable and mainstream form of identification, enabling cancer patients to benefit from early identification and more dependable and less invasive results (**Table 3**).

Despite our efforts, there are some inherent limitations to this study. A large class imbalance was present in this study, with 20 normal samples compared to 537 cancer samples. While this was accounted for during machine learning by implementing SMOTE, this imbalance must still be considered in the application of our results. While necessary, the use of SMOTE to oversample normal samples may have introduced unintended biases, primarily to the minority class (here, the normal samples) (28, 29). Our study also only tested models trained on a single miRNA rather than a multi-featured SVM model, and it exclusively used SVM rather than other machine learning algorithms such as random forest or logistic regression. This limits the confidence with which we can quantify how effective let-7 miRNAs are as biomarkers, and future studies would be needed to identify which model best predicts diagnosis based on let-7 expression.

The cohort used was also limited to TCGA. While this resulted in uniformity and reduced the risk of confounding factors, it is a limitation that must be taken into account, as any unintended biases in TCGA would be reflected in our data and could potentially compromise our results. In addition, the SVM model was tested on samples from the same cohort on which it was trained. We therefore did not use an independent cohort to assess the generalizability of the model, and any metrics obtained are only directly applicable to this cohort.

Furthermore, since this machine learning model was trained on data sourced from kidney tissues, real-world implementation directly using our or similar models would still require a biopsy to obtain the input sample, undermining the goal of a less-invasive approach. Our study suggests that let-7b is likely the best single biomarker for ccRCC in kidney tissues; however, for real-world implementation, future studies should retrain the model using urine supernatant to determine whether our findings remain consistent and which model performs best. Such a model would be better suited for non-invasive diagnosis. The diagnostic potential of let-7b is supported by both the observed expression data and the machine learning model's performance, suggesting that let-7 expression could in the future serve as a tool for determining whether a patient has ccRCC. However, let-7b may also be upregulated in other RCC subtypes, which would limit the specificity of individual biomarkers for a definitive diagnosis, highlighting the need to further investigate the use of multiple biomarkers in combination. Further research should be done to determine how specific this

behaviour is to let-7b compared to other RCC cancers and kidney cancers as a whole. The merit of this diagnosis only presents itself if let-7b's behaviour is highly dependent on cancer, allowing for models to be developed to differentiate between them. One potential implementation of diagnosis based on urine supernatant is outlined below.

First, urine samples are taken, then RNA, including miRNA, is isolated with RNA extraction kits. The RNA can be cleaned and using a miRNA-specific microarray chip that includes let-7b probes, the expression of let-7 can be quantified (11, 30). Using a machine learning model, the probability of ccRCC can be given. With this method, the presence of ccRCC can be identified earlier and non-invasively, potentially improving patient outcomes. While the proposed method is theoretically viable, further optimization and validation studies are likely to yield more effective diagnostic approaches. In the future, work could be done to put the method above into practice and check the results.

In conclusion, let-7b's average expression between normal and cancerous samples more than doubled, which is unexpected as let-7b usually acts as a tumor suppressor. However, this dramatic increase allows it to be a promising biomarker candidate, and to show this, the SVM model thus created had 90.7% accuracy. In the future, more thorough models, non-invasive methods for getting let-7 expression from patients, and more research on the unusual behaviour of let-7b's expression are needed. When these questions are answered, the role of let-7 will be better understood and ccRCC can be better identified and prevented.

MATERIALS AND METHODS

Data Acquisition

The GDC was used to obtain the data for this study. From the platform, the ccRCC data were downloaded. The miRNA levels of 557 patients were downloaded: 20 normal tissues and 537 cancerous, spread across Stage 1 to Stage 4. The data were filtered to include transcriptome profiling miRNA-Seq data, specifically miRNA Expression Quantification data. The data used were open access and were filtered to include samples whose primary site was the kidney and the diagnosis was ccRCC.

Once all the cancer files were downloaded, the filter was switched to "normal", thus including only healthy solid kidney tissue as controls. The 20 normal cases were downloaded, and SMOTE would later be used to oversample the data due to limited variability between the cases.

The organization of the folders is detailed in the appendix. This organization is necessary, as it allowed for an efficient way to structure the spreadsheet and acquire data from the files, such as a file's universally unique identifier (FileUUID) and the ID of the case, or CaseID. When rgrep or grep -r obtained the instance from the file, it also returned the path it took. By renaming the files in this way, when rgrep returned the path it took, all relevant information in the file can be obtained, facilitating data compilation.

Data Extraction

Terminal's grep functionality was used to extract the line with the target miRNA from the .txt files with the expression data. Grep -r recursively checks for the target string in any text files. This means that the command grep -r "has-let-7a"* can be run to get information from all the samples for let-7a, allowing for easy retrieval.

For each let-7 member, the text output from the command was copied and pasted into the Sublime Text editor. This allowed for formatting the text so that it could be pasted into the spreadsheet. In the spreadsheet, each sheet corresponds to a let-7 family member. The spreadsheet showed the stage of the sample (normal or cancer stage). It also showed the CaseID and the ID of the person from whom the sample was taken. Inputting this into the GDC returns all the files related to that person, including the miRNA expression from the kidneys. The FileUUID, when entered into the GDC portal, provides access to the file from which the data was taken. The spreadsheet also contains the name and URL of the file for easy access.

Data Analysis

For every miRNA sheet, two graphs were made. One was a dot plot of RPM vs. Stage for every sample. The other was a bar graph of the mean RPM vs. Stage (**Appendix**). Using an independent two-sample t-test, the p-value for each miRNA was obtained, and it was found that let-7b had the smallest p-value.

SVM Model Construction

We used the Pycharm IDE by JetBrains to code the SVM model. The SVC class from the scikit-learn library was used to develop an SVM classification model. Then, using the pandas library, key data columns—including Stage and RPM—were extracted from the spreadsheet. Normal samples were oversampled using SMOTE to address the imbalance in sample sizes.

SMOTE is a popular method for oversampling datasets when limited data is available (28). It was selected because its assumptions aligned well with the characteristics of the dataset. The data was relatively large (~500 samples), one-dimensional (RPM only), and the classes were well separated for let-7b (shown by the low p-value). All of these characteristics point to choosing SMOTE as the oversampling of choice (28). Since SMOTE does not perform well with regression problems, it was suitable here because the problem being addressed was classification (28). With no significant drawbacks, SMOTE was chosen as the best approach for oversampling.

After applying SMOTE, the data were split into training and testing sets, and the model was trained accordingly. The trained model was saved as a .joblib file to optimize efficiency, enabling reuse without repeated computation. Predictions were made using .predict and .predict_proba, which provided both class predictions and associated confidence probabilities.

Possible Errors in Application of Methodology

Possible errors could have arisen during data extraction with grep, leading to misidentified or missed miRNA entries. Misnaming or disorganizing files, such as CaseID mismatches, could have referenced incorrect samples. Additionally, manual data entry into spreadsheets is prone to typographical errors, and mistakes in statistical calculations, such as p-value determination or graphing, could misrepresent findings. As such, efforts were made to ensure that all statistical results were legitimate.

Double-checking procedures were implemented to minimize these potential errors, and outliers were validated by cross-referencing sample data with RPM values. File organization and naming conventions were carefully verified to ensure accurate data matching. Human and technical errors were reduced by inspecting statistical results for anomalies and checking data transfer processes. Future improvements could involve directly importing .txt files into the spreadsheet to eliminate manual transfer errors and leveraging additional datasets to enhance model accuracy and robustness.

A better way to avoid these errors is to directly input the .txt files with the RPM values into the spreadsheet, avoiding any transfer errors. Using more samples from websites other than the GDC would further improve the accuracy of the model and establish more concrete trends between normal and cancer expression levels for let-7b.

Another limitation of our study is that there is a possibility of unknown factors affecting the expression of let-7b. As these factors are not controlled, they could result in bias or skewing in the data. As such, the limited knowledge on let-7b is a limiting factor of this research.

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542 **Figures:**

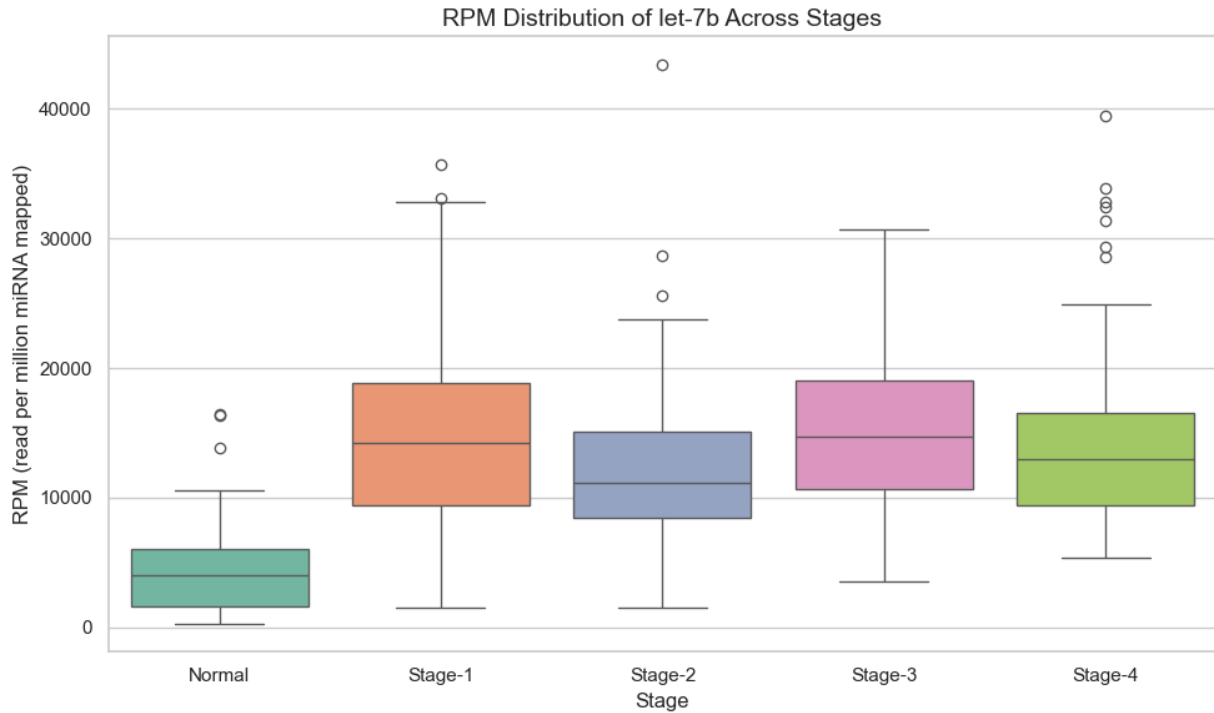
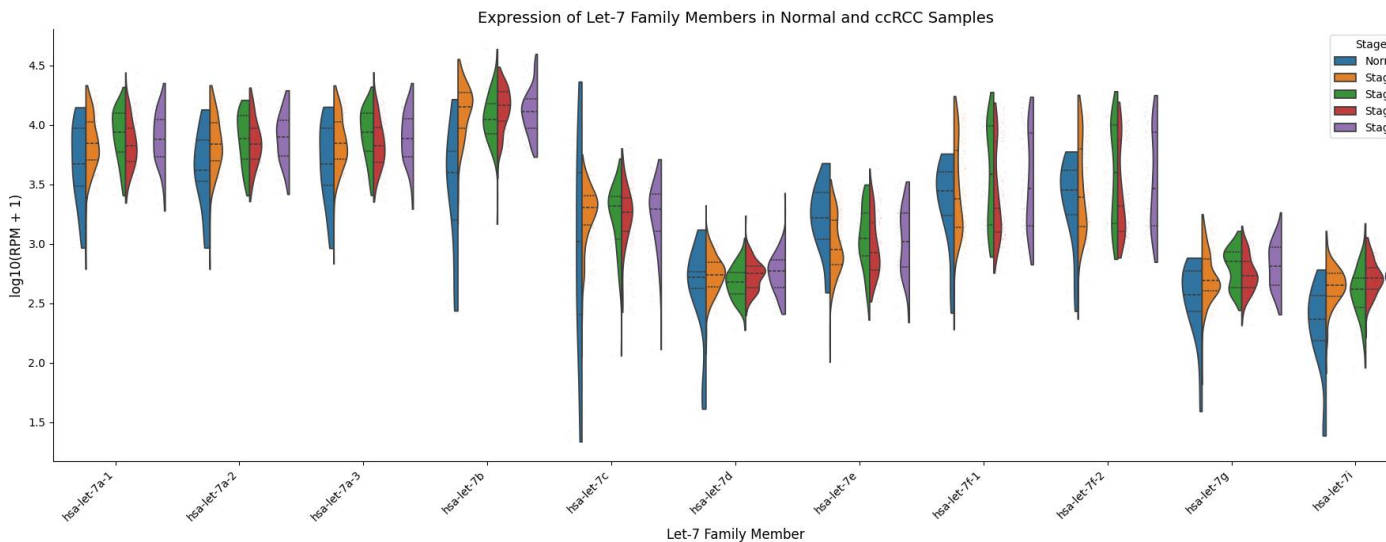


Figure 1. Box plot of mean RPM of let-7b at different cancer stages. The number of samples for each group: Normal samples: 20, Stage 1: 242, Stage 2: 65, Stage 3: 130, Stage 4: 100. It can be seen that the Normal samples have lower variance compared to cancer samples, which have larger error bars. Outliers are denoted by hollow circles, and it can be seen that all are positive outliers, meaning that they have larger expression than the mean.



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554 **Figure 2. Expression levels of let-7 in normal kidney tissue and across the four stages**
 555 **of ccRCC.** In this violin plot, normalized expression levels [$\log_{10}(\text{RPM} + 1)$] were graphed for
 556 11 let-7 family members in normal tissue and across the stages of ccRCC samples from
 557 Stages 1–4. The width of each violin represents the density of samples at a given expression
 558 level. Several let-7 family members, including let-7a isoforms, let-7b, let-7f-1, let-7f-2, let-7g,
 559 and let-7i, exhibit higher expression in tumor samples compared with normal tissue. Let-7b
 560 demonstrates one of the most pronounced increases across tumor stages. Let-7d exhibited
 561 a minimal change in regulation, while let-7c and let-7e were lowered in ccRCC. Overall,
 562 expression patterns suggest differential regulation of let-7 family members during ccRCC
 563 progression.

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565 **Tables:**

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	Stage 1	Stage 2	Stage 3	Stage 4
Normal	2.74 ($p < 0.001$)*	2.40 ($p < 0.001$)*	2.80 ($p < 0.001$)*	2.61 ($p < 0.001$)*
Stage 1		0.88 ($p = 0.055$)*	1.02 (0.675)	0.95 (0.387)
Stage 2			1.16 (0.034)*	1.09 (0.291)
Stage 3				0.93 (0.251)

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570 **Table 1. Fold changes for let-7b expression between normal tissue and cancer stages**
 571 **1–4.** This table shows fold changes between normal tissue and each cancer stage with the
 572 fold change being represented outside the brackets. Significant fold changes are denoted by
 573 an asterisk. Values over the threshold indicate no statistical significance. Bolded values
 574 highlight statistical significance from one stage to the next and can be used to assess how
 575 strongly let-7b varies with ccRCC linearly.

576

Metric	Value
Accuracy	0.907

Precision	0.887
Recall	0.936
ROC-AUC	0.931

Table 2. Overall performance metrics on SVM model. This table shows a brief overview of key diagnostics of the SVM model. Accuracy measures the proportion of total predictions that the model got right over all predictions the model made. Precision determines out of all instances that were predicted as positive and were correctly classified. Recall determines out of all actual positive instances, what proportion the model was able to catch. Receiver Operating Characteristic – Area Under Curve, or ROC-AUC, quantifies the model’s overall ability to segregate the classes.

	Predicted Normal	Predicted Cancer
Actually Normal	93	13
Actually Cancer	7	102

Table 3. Confusion Matrix of the SVM model with correct and incorrect predictions. This table details the number of false positives (13), false negatives (7), true positives (102), and true negatives (93). False positives are when the SVM predicts that the result is positive, when it is in fact negative. False negatives are when the SVM predicts that the result is negative, but it is actually positive. True positives and true negatives are when the model correctly predicts the outcome. The information in this table can then be used to determine statistics such as accuracy and precision of the SVM model.

Class	Precision	Recall	F1-Score	Support
Normal	0.93	0.88	0.90	106
Cancer	0.89	0.94	0.91	109
Accuracy	0.91	215		

Macro Avg	0.91	0.91	0.91	215
Weighted Avg	0.91	0.91	0.91	215

Table 3. Classification report with specific results from the SVM model. This table gives the precision, recall, F1 score, and support, separated by class. Precision measures the proportion of true positives to all positives predicted. Accuracy is the ratio of correct predictions to total predictions. F1 score is the harmonic mean of precision and recall, able to encapsulate both values in a clean metric. Support refers to the number of samples in each class.

APPENDIX

This is the hyperlink for the entire sheet of the let-7 members used:

https://docs.google.com/spreadsheets/d/1OxaipNosENnGiObiOAnu9GqviznvuBl8RHOFIvrl7Ro/edit?usp=drive_link.

File Organization:

FULL_DATA

- ↳ Stages (Ex, Normal, Stage 1, Stage 2, etc.)

- ↳ Unzipped the folder of the sample, renamed to CaseID

- ↳ A folder containing .txt, named after the FileUUID column

- ↳ .txt file with the expression of the miRNAs, the

name is FileName

The code for the model can be accessed on Github using the following link:

https://github.com/IvanR2625/ccrcc_svm_model.